CSC-499/299
Teaching Functional Thinking
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A Prototype "Born Accessible" Program State Visualizer

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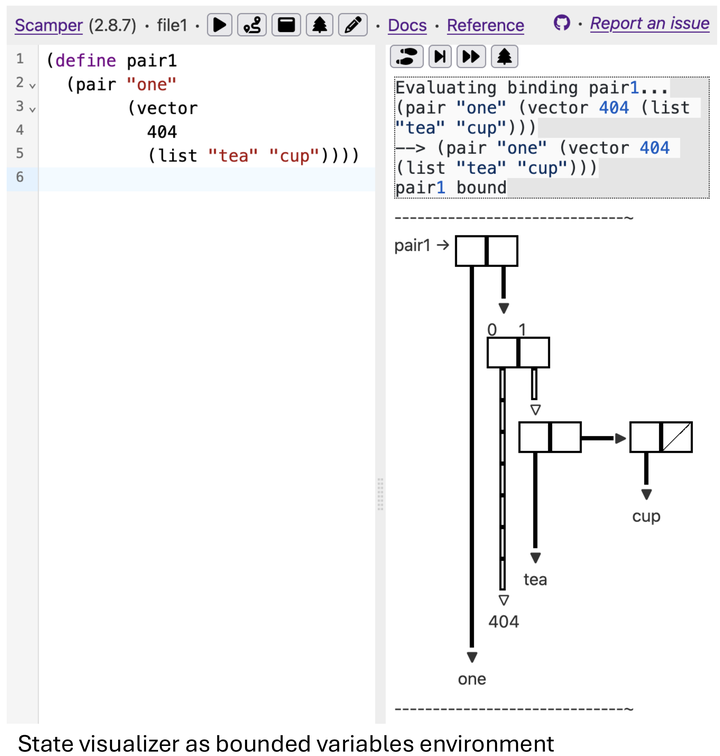
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Introduction

Students often struggle to understand the programs they write. Developing mental models of the program state is key to reducing this struggle, enabling students to more easily develop, interpret, and debug their code. This can be facilitated through program visualizers, but unfortunately, not all students have access to this technology as commonly these tools have little or poor support for blind and low vision (BLV) students. This prototype capitalizes on the lack of support by designing a program state visualizer for introductory CS students, with built-in accessibility features targeting BLV students.

Designing Visualizations

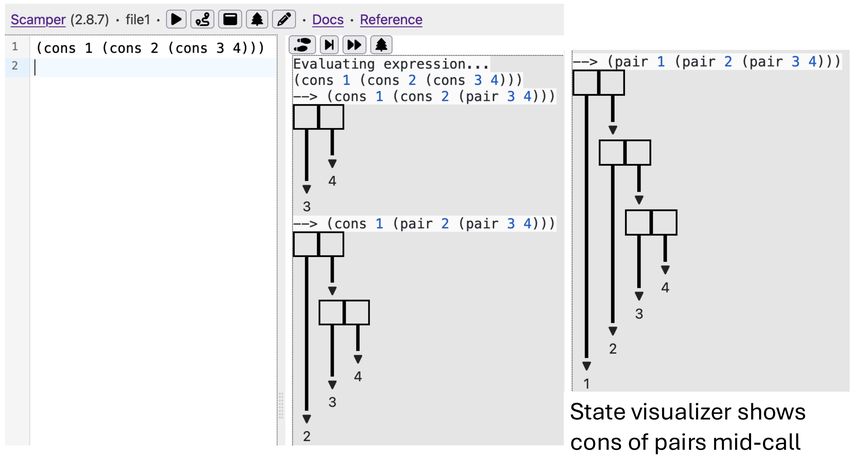
Below are some Scheme structures commonly seen in introductory computer science courses supported by the visualizer: vectors, lists, pairs, and structs. Later prototypes use dashed arrows represent mutable elements (e.g. vectors)



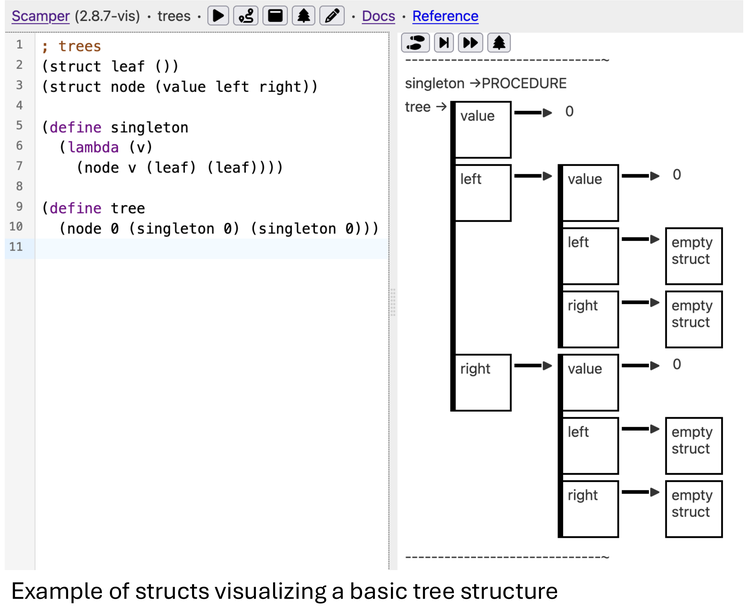
State visualizer as bounded variables environment

Visualizations in Scamper IDE

Scamper—a dialect of the functional programming language Scheme designed to teach functional programming in introductory-level computer science classes—was expanded as a part of the broader Scamper Accessibility Project being undertaken in our MAP. Scamper's IDE is web-based, with a code editor, output pane, and file browser. Scamper supports a 'trace' feature that can step through the binding and evaluation of a program, where this model of computation can be seen in the right pane. After every new bound, an environment is printed out to show the new state of the bounded variables. In addition to the bounded structures, the tracing algorithm is accompanied by a visual whenever a vector, list, or pair is found within a multitude of steps.



State visualizer shows cons of pairs mid-call



Example of structs visualizing a basic tree structure

Accessibility

We focused on improved support for neurodivergent (ND) students when creating our visualizations. To this end, we designed the visualizations to be easy to read and interpret to avoid distracting students. Each element of these visualizations includes sonification (sound that contains information about data) describing the element, to allow blind or low vision (BLV) students to benefit from them as well.

Sonification

Vectors: "vector index *number* contains *element*."

Lists: "list pair *number*, first element contains [another list/a vector/*element*]."

or "list pair *number*, second element contains [a list pair/null]."

Pairs: "non-list pair first/second element contains *element*."

Structs: "node *name* contains [a list/vector/pair or *element*]"

Navigation

A BLV student may navigate through the elements of the visualizations by pressing tab, taking the user to the next element in the page's tab order. They may also use the arrow keys to navigate within visualization squares, which allows for directional navigation of the elements. Ctrl + J jumps from the current visualization to the next.

Acknowledgments

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Selected References

- Peter-Michael Osera. [n. d.]. Scamper: A mini-Scheme for teaching. https://scamper.cs.grinnell.edu

- Jake Bell, Anna Deschamps, Eva Kapoor, Salyan Karki, Julian Kim, Nicole Moreno Gonzalez, William Pitchford, Elene Sturua, Charlotte Wade, and Sam Rebelsky. 2026. Enhancing Accessibility of a Web-based Scheme IDE to Improve Inclusion in Computing. Submitted to *57th ACM Technical Symposium on Computer Science Education* (St. Louis, Missouri, USA) (SIGCSE TS 2026).